

## ECON\*6020 Macroeconomics Theory I Fall 2026 - ASSIGNMENT 1

Due Date: Monday September 21<sup>st</sup> @ 4:00pm (start of class)

To complete this assignment, as well as other work assigned in this course, you will need access to a computer that has the Matlab or Octave software installed on it.

- MATrix LABoratory: If you wish to acquire a copy of MATLAB for your personal computer this can be done online from the MathWorks store @ <http://www.mathworks.com/> . Be sure to select the student version - MATLAB & Simulink Student Suite Version. The cost is only \$119 (USD) for an annual license. The software can be downloaded and installed right away.
- Octave: is open-source software that like MATLAB is used for matrix algebra, data analysis and computation. It uses a language that is compatible with Matlab and is therefore an alternative for MATLAB as this is not open-source. An advantage that Octave has is that it is free. If you wish to acquire a copy of Octave for your personal computer this can be done online from <https://octave.org/> .

Next, visit the Lang's Business Data Centre (BDC) website to view some of the business, economics and finance data that is available to UoG students:

- [Business, Economics and Finance Datasets](#)

For this assignment you will be required to obtain quarterly time series data from 1976Q1 to 2026Q2 for the Canadian economy. You will likely want to save this data to a spreadsheet. Select the CANSIM link at the BDC. The Canadian Socio-economic Information Management System (CANSIM) is Statistics Canada's computerized data base and information retrieval service. In my experience the Time Series View works well. Download the following series:

- Nominal and Real GDP and its components – Expenditure Based

Table 36100104 - Gross domestic product, expenditure-based, quarterly. Note that there are 31 macroeconomic variables in this table. Obtain 2 sets of data for every series in this table using the table dimensions link. The first set of data should be in current prices and seasonally adjusted at annual rates. The second set of data should be in chained 2017 prices and seasonally adjusted at annual rates.

- Aggregate hours worked

Series V4391505 - Total actual hours worked, all industries, Canada. Note that this series is provided monthly only, and therefore you will need to add items together to obtain a quarterly series.

- Population

Series v1 - Estimates of population, Canada, quarterly

- Exchange Rate – Canada-United States dollar noon spot rate. Note these series provide monthly averages. Obtain quarterly estimates by averaging the monthly statistics.
  - Series v37426 – for period 1976-2016
  - Series v111666275 – for period 2017-present

1. Now that you have the data, complete the following table:

Quarterly Canadian Business Cycle Statistics, 1976Q1-2026Q2 <sup>a</sup>

| Variable                      | % Standard<br>Deviation | Correlation with<br>Output |
|-------------------------------|-------------------------|----------------------------|
| Output                        |                         |                            |
| Consumption                   |                         |                            |
| Investment                    |                         |                            |
| Gov't Consumption Expenditure |                         |                            |
| Gov't Investment Expenditure  |                         |                            |
| Exports                       |                         |                            |
| Imports                       |                         |                            |
| Net Exports                   |                         |                            |
| Hours Worked                  |                         |                            |
| Labour Productivity           |                         |                            |
| Capital <sup>b</sup>          |                         |                            |
| Capital Productivity          |                         |                            |
| Nominal Exchange Rate         |                         |                            |
| Price                         |                         |                            |

<sup>a</sup> Output refers to gross domestic product; consumption to personal expenditures on consumer non-durables, semi-durables and services; investment to business gross fixed investment in nonresidential structures, machinery and equipment; hours worked to actual hours at work, nominal exchange rate to Canadian cents per U.S. dollar, and price to the GDP deflator. Net Exports equals one plus the ratio of net exports to GDP. Labour productivity is output divided by hours worked. Capital productivity is output divided by the capital stock. All data is from CANSIM. Before statistics were reported all series were converted to chained 2017 dollars, deflated by the population, logged and then detrended via the Hodrick-Prescott (HP) filter and using a smoothing parameter value of 1600.

<sup>b</sup> The capital stock is estimated by iterating on the following law of motion:  $K(t + 1) = I(t) + (1 - \delta)K(t)$ . The depreciation rate ( $\delta$ ) is set to 2.5% per quarter, and the initial value  $K(1976Q1)$  is set so that the average growth rates of capital and investment over the sample period are identical.

Next, calculate the correlation coefficient between hours and labour productivity:  
 $\text{corr}(\text{hours}, \text{labour productivity}) =$

2. Using the HP filter decompose the data on GDP and price into trend and business cycle components.
  - a. Graph the components of your data (in the same graph).
  - b. Identify the start of the Great Recession and the start of the recovery.
  - c. Assess how severe the Great Recession was relative to what has been experienced in the past (early 80s and early 90s).
3. Using only the information on output and price from your data, and any macroeconomic theory that you may have acquired in your previous studies, identify the general cause of the Great Recession.
4. Quantify the impact of the coronavirus pandemic on some key macroeconomic variables (output, price, hours worked).